Course Information	**Course Title**	**System Analysis**	**Course ID**	**CIE2220**
			(Course Type)	Technical Selective Sequence
	(Credit/ hours per week)	3/3	**Version (date)**	4.0 (2026. 06)
Instructor	Prof. Andrei Dragunov			

Course Learning Objective

The main objective of this course to establish fundamental concepts of system design, including representation, analysis, modeling, and construction.

	Course Objectives	Relation with Program Outcomes
1	To understand fundamental concepts of system design, including system representation and analysis.	PO 1, 3
2	To choose an appropriate system design method for a specified computer engineering application.	PO 1, 2, 3, 4
3	To analyze and design modern systems using system design methods such as the Unified Modeling Language.	PO 3, 4, 5

Course Description

This course aims students to achieve the basic concepts, principles, and techniques in system analysis and design. This course allows students to understand basic system analysis knowledge, apply it to real-world applications, and develop their own practical design.

Course Outcomes

NO	Details
1	An ability to apply knowledge of mathematics, basic science, engineering, and information technology
2	An ability to design and conduct experiments, as well as to analyze and interpret data (Analytical Experiment)
3	An ability to devise a system, component, or process to meet desired needs within realistic constraints (Ability to Design)
4	An ability to identify, formulate, and solve engineering problems
5	An ability to use techniques, skills, and engineering tools necessary for engineering practice (Tool Application)

(Prerequisites)	**None**
(Recommended Courses after This Course)	**None**
Course Software or Tool	**None**

Textbook / References

	Title	Authors	Publisher	Place	Year	ISBN
Textbook	System Analysis and Design (8th edition)	Alan Dennis, Barbara Haley Wixom. Roberta M. Roth	Wiley;		2021	**978-1119803782**
References	System Analysis and Design 12th edition	Scott Tilley	Cengage learning. Inc		2019	

Lecture type	Lecture and Q&A
(Notes)	Late homework will not be accepted

Evaluation Criteria

(Attendance)	10%	(Quizzes)	10%	(Homework Assignment)	0%	(Etc)	0%
(Mid-term Exam)	40%	(Final Exam)	40%	**(Total)**		100 %	

Methods of Evaluation	Assessment will be made on the basis of one mid-term exam, one comprehensive final examination, and 2 quizzes.

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Weekly Topical Outline of Course		
(1st Week)	Topic	Introduction to Systems Analysis and Design
	Contents	System Development Life Cycle (SDLC) overview. Introduction to AI in systems: intelligent features, data-driven requirements, AI as a system component.
(2nd Week)	Topic	Introduction to Systems Analysis and Design
	Contents	This lecture describes the evolution of system development methodologies and discusses the roles and skills required of a systems analyst. Evolution of methodologies. Role of AI-augmented analysts. Intro to object-oriented + AI-aware design.
(3rd Week)	Topic	SCRUM, Kanban. DevOps etc.
	Contents	This lecture primarily describes SCRUM and Kanban. How AI is used (and affect) in daily stand ups, sprint planning, backlog prioritization, and DevOps automation (e.g., AI Ops).
(4th Week)	Topic	Project Management
	Contents	This lecture performs a feasibility of the system; if appropriate, the system is selected and the development project begins. This lecture estimates the functionality of the project and identifies and tasks that need to be performed. Using AI for effort estimation, risk assessment, and project portfolio management.
		Quiz 1
(5th Week)	Topic	Requirements Determination
	Contents	This lecture is to determine the business requirements for a new system. This lecture begins by presenting the requirements definition, a document that lists the new system's capabilities. Lecture describes how to analyze requirements using business process automation, business process improvement and business process reengineering. Using AI (NLP) to extract and validate requirements from text or interviews.
(6th Week)	Topic	Requirements Determination
	Contents	Interviews, JAD, observation. AI-based requirement prioritization and inconsistency detection. Homework: AI vs. human requirement analysis
	Homework	the three basic steps of the analysis process the important factors in selecting an appropriate analysis strategy How does the formal system differ from the informal system? How does document analysis help you understand both?
(7th		Requirements Determination

Week)	Contents	UML use cases & activities. Using AI to generate use case diagrams from natural language descriptions.
(8th Week)	Topic	Mid-term
	Contents	Mid-term
	Exam	

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(9th Week)	Topic	Business Process and Functional Modeling
	Contents	As-is and to-be modeling. Using AI for process mining and automated workflow suggestion.
(10th Week)	Topic	Structural Modeling
	Contents	CRC cards, class diagrams. Using AI to suggest class hierarchies from requirement documents.
(11th Week)	Topic	Behavioral Modeling
	Contents	Sequence, communication, state machine diagrams. Using AI to detect missing interactions and generate test cases from behavior models.
(12th Week)	Topic	System Design
	Contents	Transition from analysis to design. Designing systems with embedded AI modules (e.g., recommendation, classification).
(13th Week)	Topic	Architecture
	Contents	Hardware/software/nonfunctional requirements. Designing AI-capable architectures (edge, cloud, hybrid). Security & ethics of AI systems.
		Quiz 2
(14th Week)	Topic	Development
	Contents	Programming, testing, documentation. AI-assisted testing (test case generation, bug prediction). Using AI to generate documentation. Optional: JIRA with AI
(15th	Topic	Final exam

Week)	Contents	Comprehensive exam covering traditional + AI-integrated system analysis.
	Exam	